MUKKESH K S

Firmware & Embedded Systems Engineer

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github.com/Mukkesh-KS | Portfolio: https://mukkesh-ks.github.io/Portfolio/

EDUCATION

SSN College of Engineering
Bachelor of Engineering (B.E.)
Electronics and Communication
2023 – 2027

TECHNICAL SKILLS

Programming: C, C++, Python, Java

Embedded Systems: STM32, ESP32, ESP32-CAM, Arduino, Raspberry Pi

Protocols: UART, SPI, I2C, CAN, RS485, LoRa

Tools: KiCad, STM32CubeIDE, VS Code, Git, GitHub, MATLAB

Concepts: RTOS/FreeRTOS, Sensor Interfacing, Hardware Bring-up, PCB Design, Motor Control, Data Acquisition Systems

PUBLICATIONS

First Author — NCICT'26

Intelligent Edge Animal Detection and Road Hazard System using STM32, ESP32-CAM, LoRa.

Presented at the 11th National Conference on ICT.

Key Research Contributions:

- Optimized TinyCNN models under tight memory constraints on ESP32-CAM.
- Engineered a long-range, low-power, distributed LoRa mesh network topology.
- Resolved safety-critical firmware latency loops for high-speed edge warnings.

CERTIFICATIONS

- NPTEL — Fuzzy Sets
- NPTEL — Waste Water Management
- AI in Engineering
- Nano to Angstrom — IC Design
- Conference Presenter — NCICT'26
- TATA Elxsi Season 3

LEADERSHIP

- Treasurer, AECE (2026–2027)
- Crowd Deputy Head, Instincts '26
- Registration Head, Invente '25
- Department Admin, Invente '25
- Decor Head, Invente '25

PROFESSIONAL SUMMARY

Electronics and Communication Engineering student and Firmware Intern with hands-on experience in embedded systems, firmware development, PCB design, and industrial communication protocols (UART, SPI, I2C, CAN, RS485, LoRa). Experienced in building end-to-end embedded solutions — from schematic design and PCB layout to firmware development, debugging, and post-fabrication hardware rework.

EXPERIENCE

Firmware Intern — Welkinrim Technologies
May 2026 – August 2026

- **System Architecture & Automation:** Architected application-level firmware for a custom STM32F767-based dual-motor thrust stand, engineering software-configurable PWM generation to simultaneously control two motors via a percentage-based throttle interface, with PWM input monitoring and a virtual COM (USB CDC) link for real-time control from a companion app — fully automating ESC calibration and test workflows.
- **Post-Fabrication Diagnostics & Rework:** Led physical board bring-up and salvaged the initial prototype batch via minimally invasive rework (track cuts, jumper wires), bypassing critical manufacturing blockers without a PCB respin.
- **Root-Cause Fault Resolution:** Prevented fatal MCU failure by diagnosing an overvoltage misrouting on the STM32's 1.2V VCAP2 pin; resolved communication and power faults caused by CAN RX/TX swaps, inverted LDO pads, and USB VBUS/VDD misconfigurations.
- **High-Frequency Data Acquisition:** Designed a deterministic data logging pipeline over an 8-channel RS485 network, sampling thrust, torque, ERPM, current, voltage, vibration, and temperature at 100 ms (10 Hz) intervals and timestamping via a battery-backed RTC that persists across power cycles, with data streamed to onboard SD storage.
- **Complex Peripheral Integration:** Built drivers and unified a high-density communication stack for real-time data transfer across 3 independent CAN buses, SPI, I2C, USB-C, Bluetooth, and Wi-Fi.

PROJECTS

LoRa-Based Roadside Animal Hazard Alert System
STM32 • ESP32-CAM • LoRa • FreeRTOS • TinyML

- Designed and developed a distributed embedded architecture consisting of multiple roadside intelligent nodes.
- Implemented TinyCNN-based TinyML inference on ESP32-CAM for real-time animal detection directly on edge devices.
- Developed FreeRTOS-based task scheduling for image acquisition, motion detection, communication, and alert management.

Handsfree Electrolarynx with ANC (Final Year Project, Ongoing)
STM32 • Active Noise Cancellation • Signal Processing • Embedded Audio

- Designing what is positioned as the first fully handsfree electrolarynx device, eliminating the need for manual hand placement against the throat during speech.
- Implementing adaptive pitch control to dynamically adjust output tone for more natural-sounding speech.
- Engineering a dual-microphone active noise cancellation (ANC) system to actively cancel the device's mechanical buzz and deliver clearer voice output.

Multi-Channel PWM/ESC Control & Load Cell Data Acquisition System
Arduino Uno • ESP32-S3 • ADS131M06 • UART • PWM • CAN • RS485 • KiCad

- Designed a custom hardware shield driving 12 ESCs simultaneously, with protection circuitry for PWM signal integrity and UART-based firmware to receive, parse, and execute timestamped PWM commands.
- Designed a custom industrial-grade PCB integrating an ADS131M06 high-resolution ADC for simultaneous multi-channel load-cell acquisition, with CAN and RS485 interfaces for industrial networking and field deployment.

Smart Helmet for Rider Safety (Ongoing)
STM32 • GPS • IMU • Haptic Feedback •

- **System Design & Crash Detection:** Developing an IoT rider safety platform using an STM32 with continuous fall-detection algorithms via motion sensors.
- **Emergency Telemetry:** Integrating GPS protocols for automated coordinate transmission alongside structural

Mini Projects: Smart IV Monitoring System, Machine Fault Detection, Smart Traffic Lights